Project Title: Byte Stay - Hotel Management System

Members :

Name : E.No. : Batch

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1. Project Overview

Byte Stay is a web-based, full-stack Hotel Management System designed to streamline both the guest experience and hotel staff operations. The system features a dual-portal architecture with a secure Staff Portal for room/worker management and a Guest Portal for browsing and booking rooms.

2. Technology Stack

- **Frontend:** HTML5, JavaScript (Vanilla/Fetch API), Tailwind CSS, FontAwesome.
- **Backend:** Java 17+, Java Servlets, Apache Tomcat 10 Web Server.
- **Database:** MySQL, JDBC (Java Database Connectivity).
- **Architecture:** Client-Server model using RESTful API principles and JSON for data transfer.

3. Core Features

- **Guest Portal:** OTP-simulated login, dynamic room search based on check-in dates and categories, real-time availability viewing.
- **Staff/Manager Portal:** Role-based access (Manager vs. standard Worker), add new rooms to the database, view all room statuses, register new staff members.
- **API Integration:** Secure cross-origin resource sharing (CORS) linking the local Tomcat server to the web interface.

4. Equal Team Contribution Breakdown

To ensure a fair and efficient distribution of work, the project is divided into three distinct pillars. Each team member owns one layer of the application.

Sri Ram Charan : Frontend Developer (UI/UX & Client Logic)

Responsibilities:

- **UI/UX Design:** Maintain and enhance the `index.html` file. Apply Tailwind CSS classes to ensure the application is responsive, visually appealing (using the glassmorphism theme), and accessible.
- **DOM Manipulation & Routing:** Manage the single-page application (SPA) state, toggling between the Start Screen, Dashboards, and Login screens.
- **API Integration:** Write the asynchronous JavaScript `fetch()` calls to interact with the Java backend (e.g., fetching room data, sending new room data).
- **Client-Side Validation:** Ensure phone numbers are 10 digits, dates are valid, and forms are fully filled before submitting to the server.

Jagannath : Backend Web Developer (Servlets & API Controller)

Responsibilities:

- **Web Server Configuration:** Manage the Apache Tomcat environment and server deployment settings in IntelliJ.
- **API Endpoints:** Create and manage Java Servlets (e.g., `RoomServlet.java`, `LoginServlet.java`).
- **HTTP Request/Response Handling:** Write the `doGet` and `doPost` methods. Extract data from frontend HTTP requests and format Java data into clean JSON responses.
- **Security & Networking:** Handle CORS configurations so the frontend can securely communicate with the backend, and manage HTTP status codes (200 OK, 404 Not Found, 500 Error).

Chakradhar Rao : Database Architect & Java Logic (JDBC & Models)

Responsibilities:

- **Database Design:** Design the MySQL relational database. Create and maintain the tables (`Rooms`, `Staff`, `Guests`, `Bookings`) and their relationships (Primary/Foreign keys).
- **Java Models:** Create the Object-Oriented representations of the data (e.g., `Room.java`, `User.java` classes with appropriate getters, setters, and constructors).

- **Data Access Object (DAO):** Write the core JDBC logic (e.g., `HotelDAO.java`, `DatabaseConnection.java`). This includes writing the SQL `SELECT`, `INSERT`, `UPDATE`, and `DELETE` queries and safely passing data to and from the database.

5. Development Workflow & Collaboration

1. **Database First:** *Role 3* finalizes the table structures and provides *Role 2* with the Java Models (e.g., `Room.java`).
2. **API Construction:** *Role 2* builds the Servlets to call the DAO methods provided by *Role 3*, exposing them at URLs like http://localhost:8080/api/rooms.
3. **Frontend Binding:** *Role 1* designs the UI and binds the buttons and tables to the URLs created by *Role 2*.

6. Future Enhancements (Phase 2)

- Implementation of an AI Chatbot (Gemini API) for the Guest portal.
- PDF invoice generation for checkout.
- Password hashing (Bcrypt) for worker authentication.

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